Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2007-I-Q06] | Part B (Q6-Q10)

A frog begins at zero. Each step goes to either the next strictly larger multiple of three or the next strictly larger multiple of thirteen. Count distinct coordinate sequences ending at 39.

青蛙从零开始，每步跳到严格更大的最近三倍数或最近十三倍数。求终止于三十九的不同坐标序列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2007-II-Q06] | Part B (Q6-Q10)

A decimal digit string is parity-monotonic when every odd digit is followed by a strictly larger digit and every even digit by a strictly smaller one. Count four-digit positive integers with this property.

若十进制数字串中每个奇数字后都接更大数字、每个偶数字后都接更小数字，称其为奇偶单调。求符合的四位正整数个数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2013-II-Q09] | Part B (Q6-Q10)

Tile a seven-cell row using arbitrary positive integer-length tiles, each colored red, blue, or green. Tile boundaries distinguish tilings even when neighboring tiles share a color. Count tilings using all three colors, modulo 1000.

用正整数长度的一格宽瓷砖铺满七格行，每砖为红蓝绿之一。即使相邻砖同色，改变砖的分界仍视为不同。求三色均出现的铺法数模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2015-II-Q10] | Part B (Q6-Q10)

Count permutations $a_1, \ldots, a_7$ of $1, \ldots, 7$ satisfying $a_k \leq a_{k+1} + 2$ for all $k < 7$.

求 1 至 7 的排列 $a_1, \ldots, a_7$ 中，满足对所有 $k < 7$ 均有 $a_k \leq a_{k+1} + 2$ 的排列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-II-Q08] | Part B (Q6-Q10)

A frog starts at $(0, 0)$ and may move one or two units east or one or two units north each jump. Count ordered jump sequences ending at $(4, 4)$.

青蛙从原点出发，每次可向东一或二单位，或向北一或二单位。求到达 $(4, 4)$ 的有序跳跃序列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2024-I-Q06] | Part B (Q6-Q10)

Count shortest grid paths from $(0,0)$ to $(8,8)$ that use unit right/up steps and turn exactly four times.

在方格上从 $(0,0)$ 到 $(8,8)$，每步向右或向上一个单位。求恰好转向四次的最短路径条数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2000-I-Q15] | Part C (Q11-Q15)

A stack of 2000 distinctly labeled cards is processed by repeatedly putting its top card onto a table and moving the next card to the bottom of the stack. The table receives labels in order $1, 2, \ldots, 2000$. How many cards originally lay above label 1999?

一叠 2000 张不同标号牌反复执行：顶牌放到桌上，下一张顶牌移到牌堆底部。桌上得到的标号顺序为 $1, 2, \ldots, 2000$。原牌堆中标号 1999 上方有多少张牌？

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2001-I-Q14] | Part C (Q11-Q15)

Nineteen houses lie in a row. Choose which receive mail so that no adjacent pair both receive it and no three consecutive houses all receive none. How many patterns are possible?

十九户排成一行。选择当天收信的住户，要求相邻两户不能都收信，连续三户也不能都不收信。求可能模式数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2004-I-Q15] | Part C (Q11-Q15)

For positive integers define $f(1)=1$, $f(x)=x/10$ when $10 \mid x$, and $f(x)=x+1$ otherwise. Starting at x, let $d(x)$ count the terms up to and including the first 1. If M counts x with $d(x)=20$, find the sum of the distinct prime divisors of M.

对正整数定义 $f(1)=1$；若 $10 \mid x$，则 $f(x)=x/10$；否则 $f(x)=x+1$。从 x 开始迭代，令 $d(x)$ 为包括首次一在内的项数。若 $d(x)=20$ 的起点数为 M，求 M 的不同质因数之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2005-I-Q13] | Part C (Q11-Q15)

A lattice path uses steps $R=(1,0), U=(0,1), D=(1,1)$, but may not have adjacent RU or UR. Count paths from $(0,0)$ to $(5,5)$.

格点路径允许步长 $R=(1,0), U=(0,1), D=(1,1)$，但不得相邻出现 RU 或 UR。求从 $(0,0)$ 到 $(5,5)$ 的路径数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2006-I-Q11] | Part C (Q11-Q15)

Stack cubes of side lengths $1, \ldots, 8$, using all once. A cube of length k may have immediately above it only a cube of side at most $k+2$. Find the number of towers modulo 1000.

将边长一至八的八个立方体各用一次叠成塔，边长 k 的上面紧邻的立方体边长至多 $k+2$。求塔数除以一千的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2008-I-Q11] | Part C (Q11-Q15)

Count length-fourteen words in A, B whose maximal A runs all have even positive length and maximal B runs all have odd positive length.

求长十四的 A, B 字符串数，使每个极大连续 A 段长为正偶数，每个极大连续 B 段长为正奇数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2015-II-Q12] | Part C (Q11-Q15)

How many length-10 strings over $\{A, B\}$ have every maximal run of identical letters of length at most 3?

由 A, B 组成的长度 10 字符串中，连续相同字符的每个极大段长度都不超过 3，这样的字符串有多少个?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2016-II-Q12] | Part C (Q11-Q15)

Six labeled regions form a ring: each region shares an edge only with its two neighbors in the six-cycle. Paint them using four available colors, with neighboring regions different. Rotations are distinct because the positions on the wall are fixed. Count the colorings.

六个固定位置区域围成一环，每区仅与六循环中的左右邻区共边。用四种颜色涂色，邻区颜色不得相同；墙上位置固定，旋转后的不同位置涂法不合并。求涂法数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-I-Q14] | Part C (Q11-Q15)

A frog starts at S on the cycle with consecutive vertices $S, P_1, P_2, P_3, E, P_4, P_5$. At each non-$E$ vertex it chooses either adjacent vertex; reaching E ends the path. Count distinct paths reaching E within twelve jumps.

七边循环顶点依次为 $S, P_1, P_2, P_3, E, P_4, P_5$。青蛙从 S 出发，每次在非 E 点可跳至任一邻点，一到 E 即停止。求不超过十二步到达 E 的不同路径数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2018-II-Q11] | Part C (Q11-Q15)

Count permutations of $1, \ldots, 6$ such that for each $1 \leq k \leq 5$, the first k terms contain an element greater than k.

求一至六的排列数，使对每个 $1 \leq k \leq 5$，前 k 项中至少有一项大于 k。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2019-II-Q12] | Part C (Q11-Q15)

A progressive sequence of positive integers is strictly increasing and each term divides the next. Count all finite progressive sequences whose sum is 360, allowing length one.

正整数有限序列严格递增且每项整除下一项，称为递进序列。求项和为三百六十的此类序列数，允许只有一项。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2023-I-Q11] | Part C (Q11-Q15)

How many subsets of $\{1, \ldots, 10\}$ contain exactly one unordered pair of consecutive integers?

$\{1, \ldots, 10\}$ 的子集中，恰含一对相邻整数的共有多少个?

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2026-I-Q15] | Part C (Q11-Q15)

A cell loop is the one-cell-thick boundary of an axis-aligned $a \times b$ cell rectangle, with $a, b \geq 2$; when one dimension is two, its whole rectangle is the loop. Count partitions of a 10×10 grid into exactly five disjoint cell loops.

格子环指轴对齐 $a \times b$ 长方形（$a, b \geq 2$）的一格厚边框；若一边为二，则整个长方形都是环。求将 10×10 方格恰好分成五个互不重叠格子环的划分数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.5 Recurrences, lattice paths & tilings

递推、格路与铺砌

[AIME-2026-II-Q14] | Part C (Q11-Q15)

For integers define $a \circ b = a - b$ when a is odd and b even, and $a \circ b = a + b$ otherwise. Count compositions $a_1 + \cdots + a_j = 12$ into positive integers whose left-associated $\circ$ value is zero.

对整数定义：若 a 为奇数且 b 为偶数，则 $a \circ b = a - b$，否则 $a \circ b = a + b$。求正整数序列之和为 12 且从左至右计算的 $\circ$ 值为零的序列数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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